SẮC KÝ v2

FUTUREME / FUTURE LETTER CODEX IMPLEMENTATION GUIDE v1.0

Product rules • Trigger Engine • State Machine • Desktop UX/UI • Acceptance Contract

IMPLEMENTATION INTENT

Đây là baseline build-ready để Codex implement FutureMe Letter v2. Không được tự giản lược trigger semantics, đổi lifecycle, hoặc biến UI thành một feature email/reminder thông thường.

Module	Future Letter / FutureMe
Desktop priority	MacBook / 16:10 first
Authoring flow	Viết thư → Trang trí thư → Preview → Seal
Trigger logic	Random / Specific Date / Check-in Streak / Diary Streak / Emotion Frequency / Done Action
Boolean logic	AND / OR + comparators >, >=, =, <=, <
Decoration	Envelope + Seal asset placeholders
Color accent	Không có letter color accent
Style authority	Sắc Ký v2 canonical style guide + Home / Check-in / Diary visual language

1. Product North Star

FutureMe là một hệ thống để user viết một lá thư cho chính mình ở hiện tại, niêm phong nó, và để lá thư quay trở lại vào một ngày hoặc một trạng thái tương lai có ý nghĩa. Nó không phải scheduled note, email-to-self hay reminder.

CORE RITUAL
Write → Decorate → Preview → Seal → Wait → Trigger → Open → Reappear until Archived

- Diary ghi lại hiện tại; FutureMe gửi một thông điệp từ hiện tại tới một phiên bản tương lai của chính mình.
- Trigger engine thuộc FutureMe domain. Home chỉ render projection của thư đang available.
- Một lá thư đã mở vẫn có thể quay lại trong tương lai cho đến khi user chủ động Archive.
- Archive giữ ký ức nhưng loại thư khỏi trigger/random pool; Delete xóa sạch.
- Không có reflection sau khi mở thư. Emotional payoff nằm ở opening ritual/animation.

2. Canonical Lifecycle

```
DRAFT
  ↓
WRITING
  ↓
DECORATING
  ↓
PREVIEW
  ↓
SEALED / WAITING
  ↓ trigger qualifies or random engine selects
AVAILABLE / UNOPENED
  ↓ user opens
OPENED
  ↓ may become eligible again later
REAPPEAR / AVAILABLE AGAIN
  ↓ user explicitly archives
ARCHIVED

DELETE is allowed from any state and removes the letter completely.
```

- Draft is editable.
- After Seal, content is immutable. Không có “unseal to edit”.
- Once a conditional trigger qualifies, it latches: thư không quay về Waiting chỉ vì condition không còn đúng.
- Available-but-unopened carries over across days until opened.
- Opened today có thể vẫn nằm Home đến hết ngày; opened từ ngày trước không còn ở Home trừ khi có một trigger/random event mới đưa thư trở lại.
- Archive là trạng thái giữ lại, không phải Delete.

3. Authoring Contract

3.1 Write Letter

- Title: bắt buộc.
- Body: bắt buộc, plain multiline.
- Không rich text.
- Không prompt.

- Không media trong v1.
- Optional context: attach Emotion và/hoặc Check-in bất kỳ trong lịch sử; không cần cùng ngày.
- Attachment chỉ là context/memory của lúc viết, không tự biến thành trigger.
- Không có letter color picker / color accent.

3.2 Authoring sequence

```
STEP 1  Viết thư
STEP 2  Trang trí thư
STEP 3  Preview
STEP 4  Seal
```

LOCK
Không đổi thứ tự flow. Trigger setup có thể nằm trong bước Viết hoặc một sub-section trước Preview, nhưng user-facing ritual phải vẫn kết thúc bằng Decorate → Preview → Seal.

4. Decoration Contract

- User chọn 1 Envelope asset.
- User chọn 1 Seal asset.
- Hiện chưa có asset thật: implement bằng placeholder slots/asset IDs, không hardcode artwork tạm như production truth.
- Khi Seal, snapshot/version selected decoration vào Letter để thư cũ không đổi hình nếu catalog asset tương lai thay đổi.
- Decoration là visual identity chính của FutureMe; không thêm color accent picker để thay thế.

5. Trigger Engine - Canonical Rules

FutureMe phải hỗ trợ nhiều trigger condition và kết hợp AND/OR. Các condition không phải text labels; chúng phải trở thành structured rule objects có comparator, threshold/window và evidence khi trigger.

5.1 Random Date

CRITICAL SEMANTICS
30% là chance ở cấp ngày cho cả RANDOM POOL, không phải 30% trên từng lá thư.

```
eligible random pool = N letters

Each local day:
  if N == 0 → no random event
  else roll once for the pool

  30% → select AT MOST 1 letter
  70% → select none

Example: 10 random letters does NOT mean 10 × 30%.
The whole pool gets one 30% daily chance.
```

- Admin/config có thể điều chỉnh daily chance về sau; baseline hiện tại = 30%.
- Nếu random được chọn, tối đa 1 random letter được đưa ra trong ngày đó.
- Conditional/specific-date letters vẫn có thể trigger cùng ngày, nên tổng available letters có thể >1.
- Random selection ưu tiên letters có opened_count thấp hơn, nhưng không deterministic 100%. Dùng weighted random/fairness để tránh starvation.
- Random letter sau khi mở vẫn quay lại random pool trong tương lai cho tới khi Archived.
- Khi random event đã chọn một letter, letter latches available và carry-over đến khi user mở; không reroll nó mỗi ngày.

5.2 Specific Date

- Date only; không exact time.
- Đến ngày → trigger → available.
- Nếu user không mở đúng ngày, letter carry-over cho tới khi mở.
- Specific-date trigger event nên được evidence-log.

5.3 Check-in Streak

- Metric: current/check-in streak theo canonical Check-in streak semantics của app.
- Support comparator: >, >=, =, <=, <.
- Default suggested comparator: >=.
- Trigger latches once condition becomes true.

5.4 Diary Streak

- Metric: canonical Diary streak.
- Support comparator: >, >=, =, <=, <.
- Default suggested comparator: >=.
- Do not duplicate streak calculation inside FutureMe; call/reuse canonical Diary streak service/selector.

5.5 Emotion Frequency

WINDOW SEMANTICS
30-day chain bắt đầu từ ngày có Check-in đầu tiên của emotion đó. Đây KHÔNG phải calendar month và cũng KHÔNG phải generic rolling 30-day window.

```
First qualifying Check-in with Emotion X = D0
Window = D0 ... D0 + 29 days

Count occurrences of Emotion X inside that chain.
If count [comparator] threshold → trigger.
```

- Condition cho phép chọn Emotion X.
- Cho phép chọn scope: Primary only hoặc Primary + Secondary. Default recommended = Primary only.
- Cho phép comparator >, >=, =, <=, <.
- Threshold Y là số lần xuất hiện trong chain.
- Nếu chain kết thúc mà không đạt threshold, future implementation cần xác định cách mở chain mới từ occurrence tiếp theo; Codex phải implement deterministic rule và test. Recommended: next occurrence after prior window closes becomes new D0.
- Trigger evidence phải lưu emotion_id, scope, threshold, observed count, window_start, window_end.

5.6 Done Action

- Mode A: lifetime cumulative completed Actions.
- Mode B: completed Actions trong period/window.
- Period/window v1 nên hỗ trợ ít nhất: All time, Last 7 days, Last 30 days; custom range có thể để extension.
- Support comparator >, >=, =, <=, <.
- Reuse canonical Action completion data; không tạo counter riêng dễ drift.

6. Boolean Rule Builder

- Support nhiều conditions.
- Support AND / OR.
- Support condition groups.
- V1 không cần arbitrary infinitely nested Boolean AST UI. Một cấp Condition Group + group operator là đủ để biểu diễn (A AND B) OR C hoặc tương đương.
- Comparator control nằm ngay cạnh metric/value: >, >=, =, <=, <.
- Không dùng natural-language-only rules; phải persist structured data.

```
Example:
GROUP 1 (ALL / AND)
  Diary streak >= 7
  Emotion Bình yên >= 3 times / 30-day chain

OR

GROUP 2 (ALL / AND)
  Done Actions >= 5 / last 30 days
```

7. Trigger Latching & Evidence

- Một conditional rule chỉ cần true một lần để letter trở thành available.
- Sau đó điều kiện có mất cũng không re-lock thư.
- Persist triggered_at / available_since.
- Persist trigger evidence để explainability và audit.
- Random selection cũng tạo trigger event/evidence riêng.
- Một letter có thể có nhiều trigger events trong lifetime do reappearance.

```
TriggerEvidence
- letter_id
- event_id
- trigger_type
- rule_snapshot
- observed_values
- window_start?
- window_end?
- triggered_at
- source_ref_ids?
```

8. Conceptual Data Model

Tên field/table là conceptual. Codex phải inspect schema hiện hữu và map/reuse nếu đã có. Không tự migration phá dữ liệu chỉ để đổi tên.

```
FutureLetter
- id
- user_id
- title
- body
- state / lifecycle fields
- created_at
- updated_at
- sealed_at?
- first_opened_at?
- last_opened_at?
- opened_count
- archived_at?
- deleted_at? (only if implementation requires; product semantics = hard delete)
- envelope_asset_id?
- seal_asset_id?
- decoration_snapshot?

FutureLetterContextAttachment
- letter_id
- attachment_type: emotion | checkin
- ref_id
- snapshot?
- position

FutureLetterRuleGroup
- id
- letter_id
- group_operator: AND | OR
- position

FutureLetterRule
- id
- group_id
- rule_type
- comparator
- threshold/value
- config_json
- position

FutureLetterTriggerEvent
- id
- letter_id
- trigger_type
- rule/evidence snapshot
- available_since
- opened_at?
- created_at
```

- opened_count tăng khi user thực sự mở thư.
- first_opened_at chỉ set lần đầu.
- last_opened_at update mỗi lần mở.
- Archive loại khỏi all future eligibility/random evaluation.
- Delete xóa sạch theo product semantics.
- Draft có thể save nhiều lần; Seal freezes content + trigger + decoration snapshots.

9. Home Projection Contract

- Home không tự evaluate trigger.
- FutureMe domain/service expose available letters.
- Waiting letters không show như openable.
- Available unopened show trên Home.
- Available từ hôm trước chưa mở carry-over.
- Opened today có thể tiếp tục nằm Home tới end of local day.
- Opened earlier không nằm Home trừ khi letter re-triggers/reappears.
- Multiple available → bounded carousel.
- User click trực tiếp envelope/seal để mở; không cần CTA “Mở ngay”.

10. Opening Ritual

NO REFLECTION
Sau khi mở thư không hỏi mood, rating, note hay reflection. Interaction kết thúc bằng reveal/read experience.

- Opening animation là feature-level signature của FutureMe.
- Sequence recommended: envelope lift → seal pulse → seal break/lift → flap open → paper slide/reveal → read view.
- Duration khoảng 1.0-1.8s ở normal motion.
- Không confetti, không mascot, không excessive sparkles.
- Có reduced-motion variant.
- Animation phải cảm giác intimate/anticipatory, không cartoon.

11. Archive vs Delete

Action	Archive	Delete
Keeps content/history	Yes	No
Eligible to reappear	No	No
Visible in Archive/History	Yes	No

Current implementation assumption: Archive is conceptually reversible if the existing product already supports restore/unarchive. Nếu schema chưa có restore, Codex không được tự thêm behavior ngoài spec; report the gap.

12. Desktop UI System - Mandatory

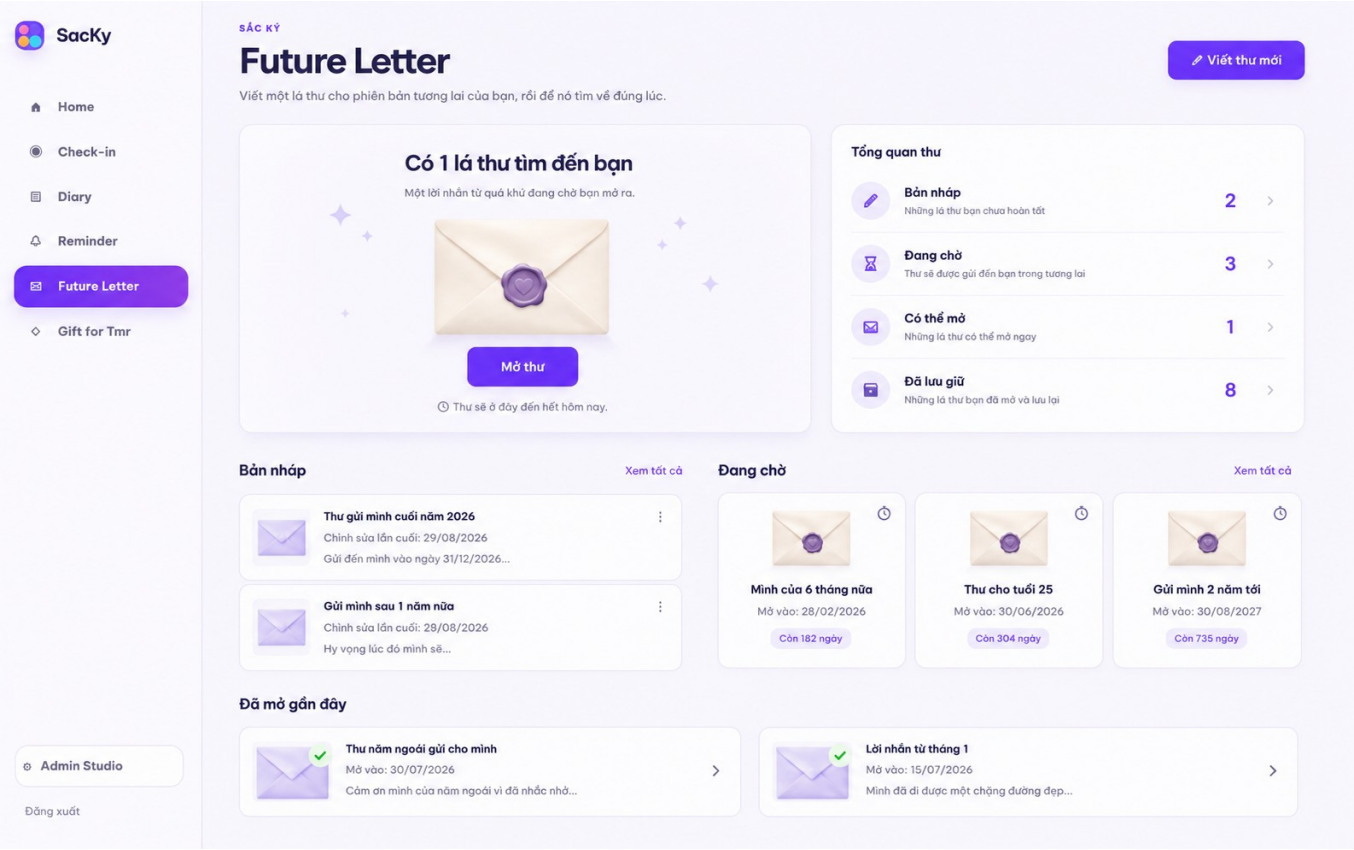
Các mockup dưới đây là desktop direction. Codex phải follow Sắc Ký v2 canonical style guide và đồng bộ trực tiếp với Home / Check-in / Diary hiện tại, không dùng storyboard/mobile styling cũ.

Canvas	#F8F6FF
Surface	#FFFFFF
Ink	#231B46
Muted	#625A75
Accent Violet	#743CFF
Accent Strong	#5326BE
Accent Soft	#ECE4FF
Border	#E5DDF5
Danger	#FF5C70
Font	Be Vietnam Pro 400 / 500 / 600 / 700

- Desktop sidebar khoảng 212px, giống app shell hiện tại.
- Content max-width / horizontal rhythm phải match Home/Diary; không tự tạo một layout center-column mới.
- Spacing first: 4/8/12/16/20/24/32/40/48/64.
- Radii: 8/12/16/22/28/pill, dùng có hệ thống.
- Space before boxes. Không card-inside-card nếu không cần.
- One visual anchor per zone.
- Không mascot/kawaii.
- Không wellness pastel cliché.
- Không SaaS dashboard dense.
- Không glassmorphism diện rộng.
- Không giant wrapper rectangle bao toàn page.
- Không native horizontal scrollbar cho carousel/rail; dùng bounded arrows.

IMPORTANT
Envelope/Seal có thể là visual anchor và có chất liệu riêng, nhưng shell, typography, grid, spacing, buttons, tabs, drawer/modal behavior phải giống Home/Check-in/Diary.

F01 - Landing / Library

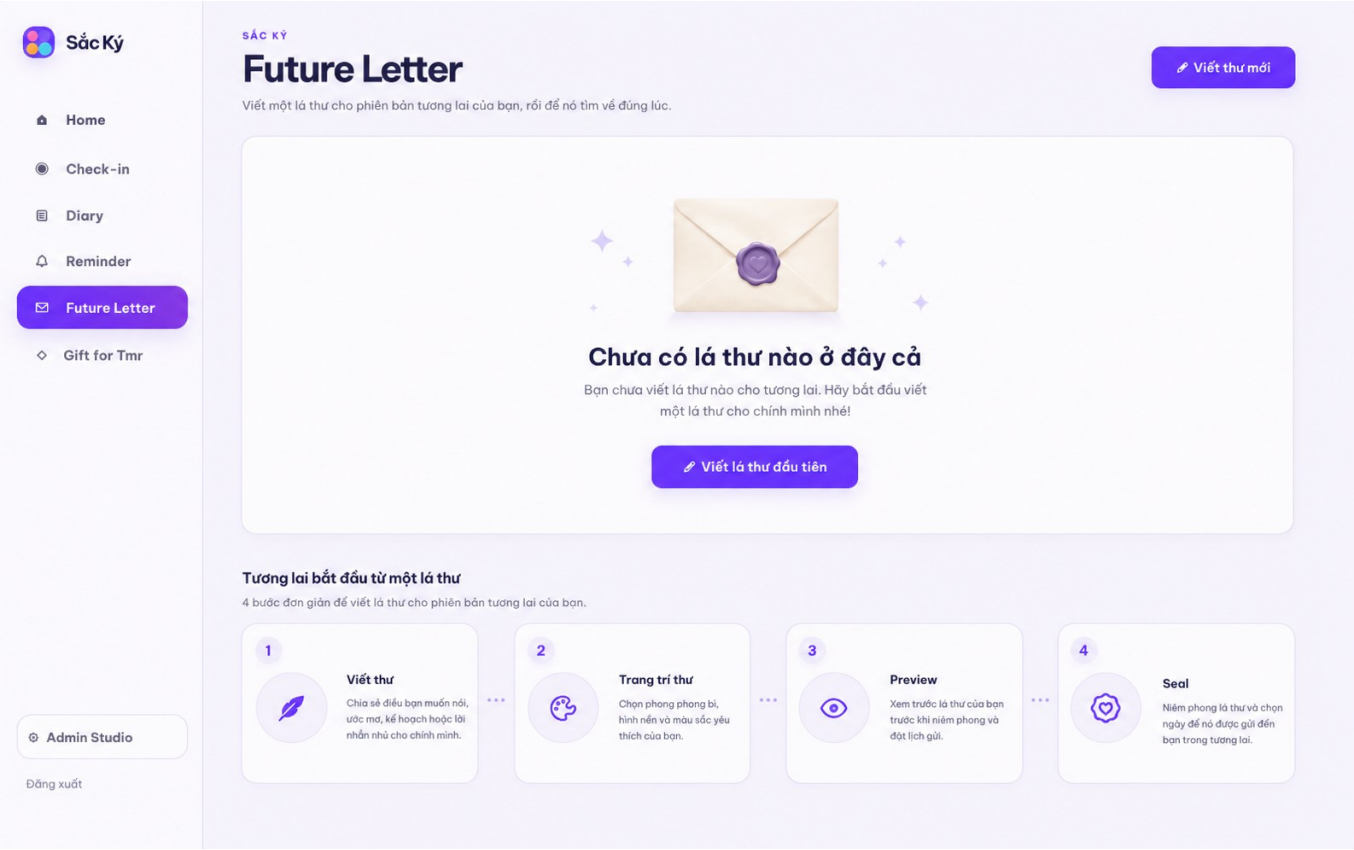


Approved desktop direction - F01 - Landing / Library

Codex implementation contract

- Hero ưu tiên available letter nếu có; click envelope/seal để mở.
- Right/summary zone cho Draft / Waiting / Available / Archived counts ở mức nhẹ.
- Sections: Draft, Waiting, Recently Opened. Không biến thành email inbox.
- CTA “Viết thư mới” dùng canonical violet button.

F02 - Empty State

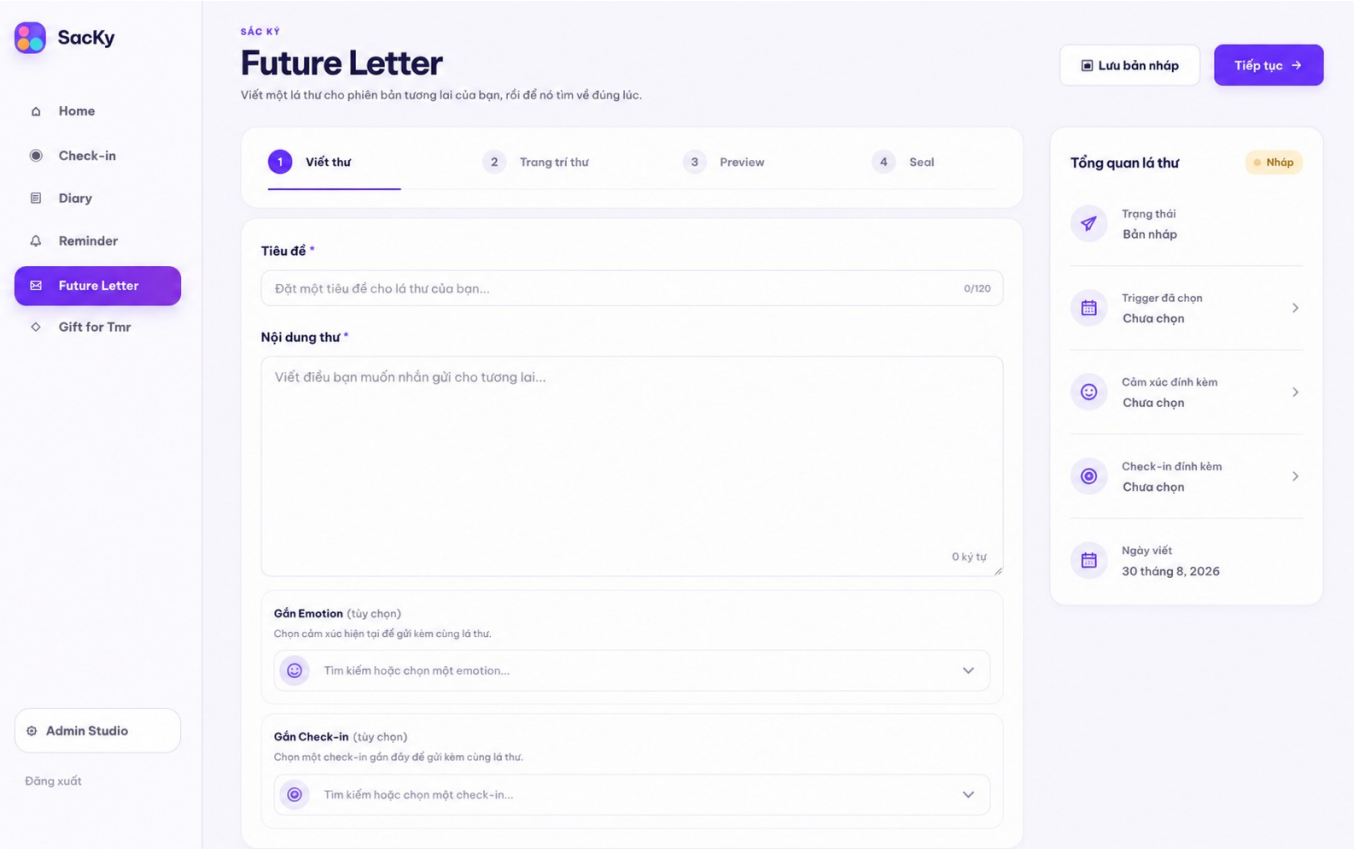


Approved desktop direction - F02 - Empty State

Codex implementation contract

- Editorial empty state, một visual anchor envelope.
- CTA viết lá thư đầu tiên.
- Có thể giải thích 4 bước Write → Decorate → Preview → Seal nhưng tránh dashboard-heavy.

F03 - Write Letter

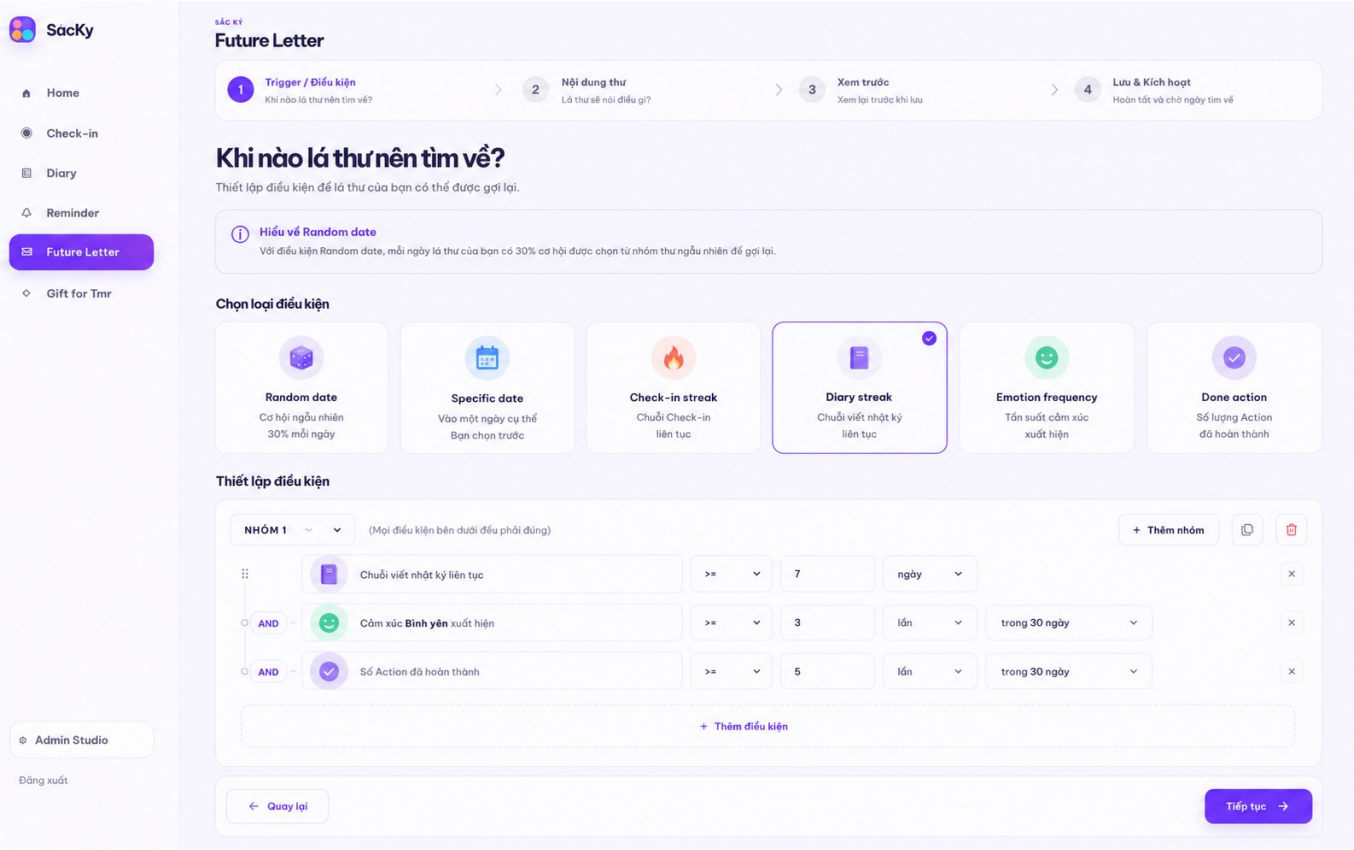


Approved desktop direction - F03 - Write Letter

Codex implementation contract

- Title + Body là core writing area.
- Optional Emotion/Check-in context.
- Save Draft + Continue.
- Right rail summary sticky theo Diary pattern.
- Không color picker, prompt, media.

F04 - Trigger Builder

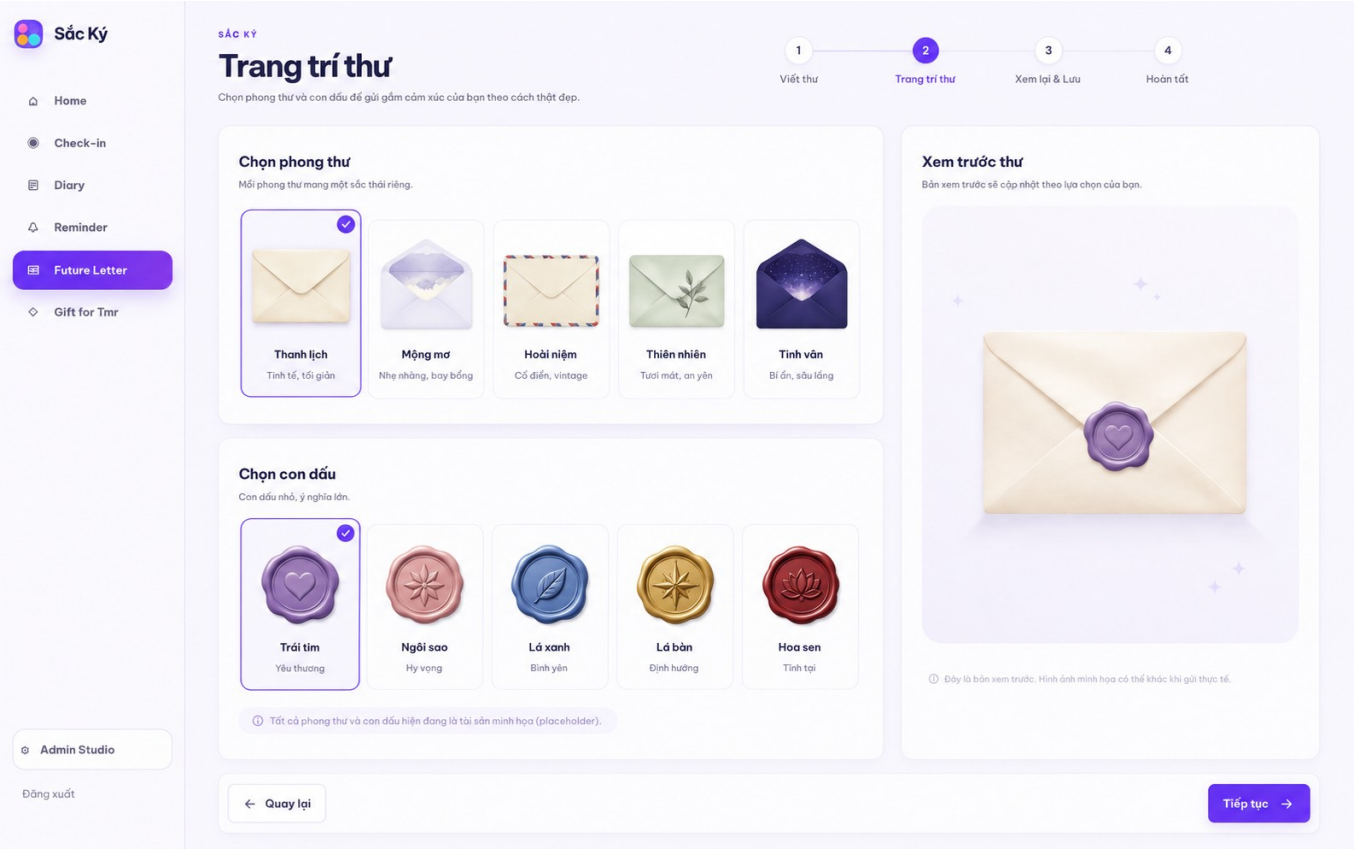


Approved desktop direction - F04 - Trigger Builder

Codex implementation contract

- Full desktop workspace vì logic AND/OR cần chiều ngang.
- Trigger type cards phía trên; structured rules phía dưới.
- Comparator controls >, >=, =, <=, < nằm cạnh value.
- Condition groups rõ ràng nhưng không giống enterprise workflow builder.
- Random semantics phải mô tả đúng 30% at pool/day level.

F05 - Decorate Letter

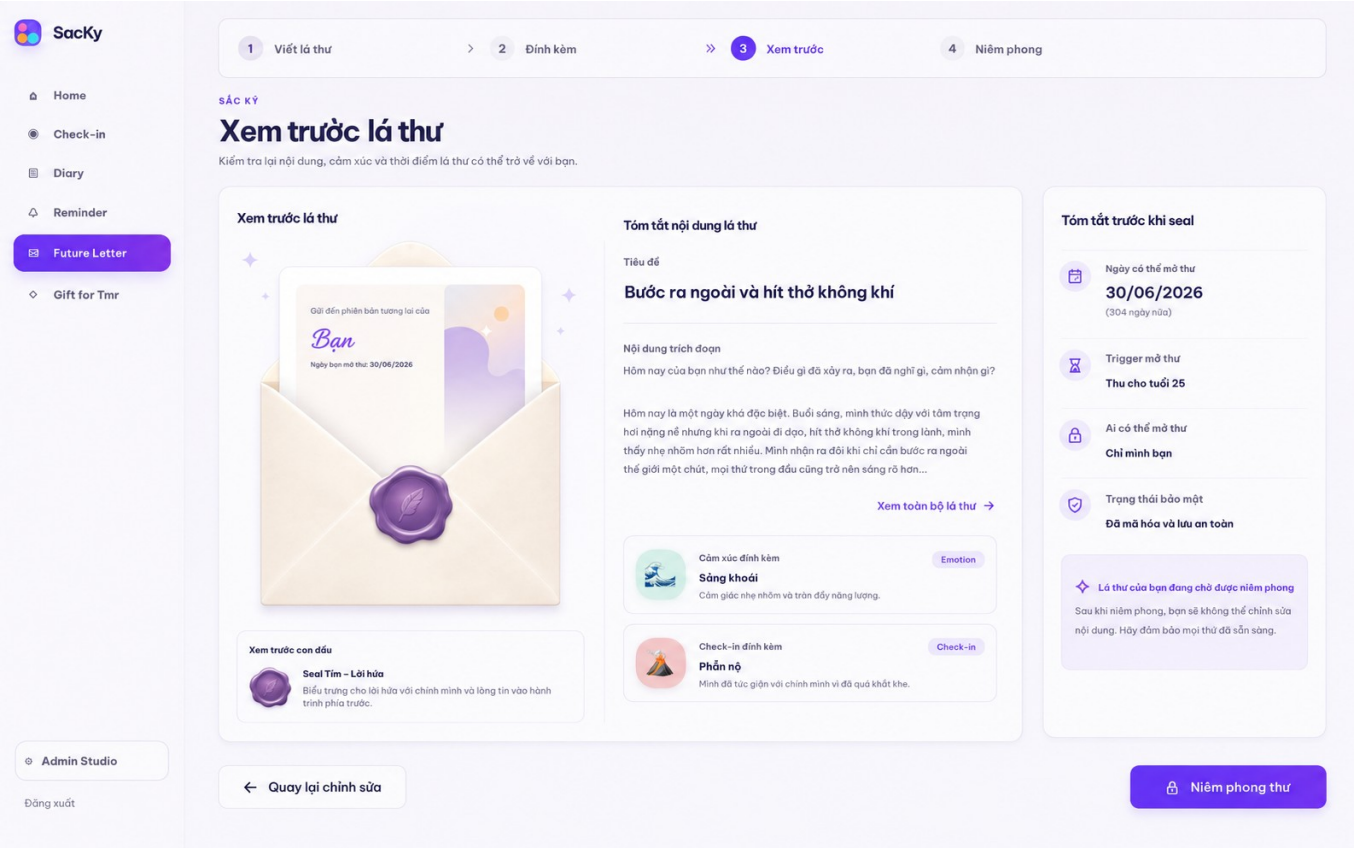


Approved desktop direction - F05 - Decorate Letter

Codex implementation contract

- Envelope catalog + Seal catalog.
- Assets hiện là placeholder; preserve slot structure for later asset mapping.
- Large live preview bên phải.
- No letter color accent.

F06 - Preview / Seal Readiness

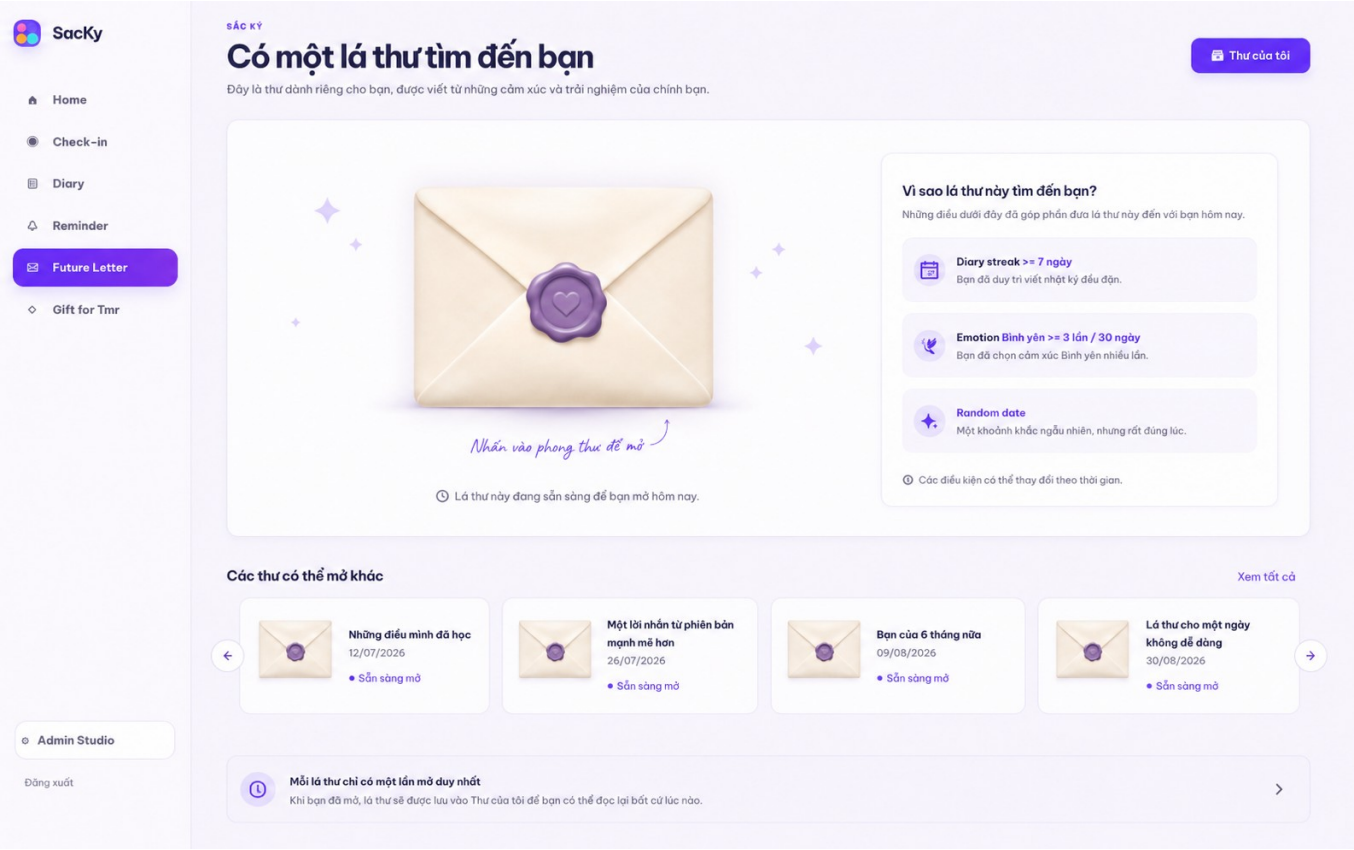


Approved desktop direction - F06 - Preview / Seal Readiness

Codex implementation contract

- Preview title/body/context/trigger summary/decoration.
- Quay lại chỉnh sửa trước Seal.
- Primary CTA Niêm phong thư.
- Sau Seal content immutable.

F07 - Waiting / Sealed Library

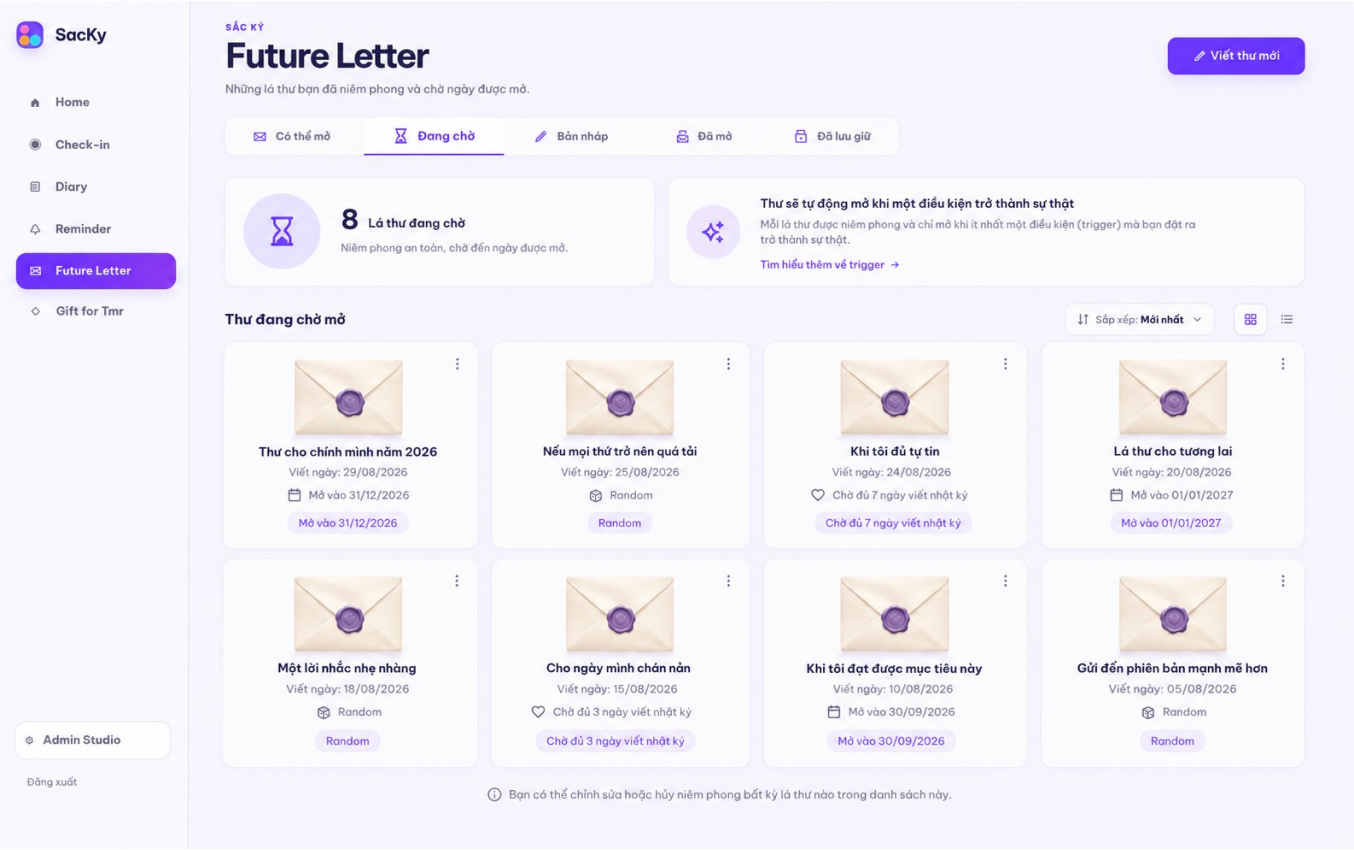


Approved desktop direction - F07 - Waiting / Sealed Library

Codex implementation contract

- Grid/list of sealed waiting envelopes.
- Show only safe metadata: title, written date, trigger summary.
- No Edit after Seal.
- Actions limited to inspect/delete/archive only if applicable by state.

F08 - Available Letter

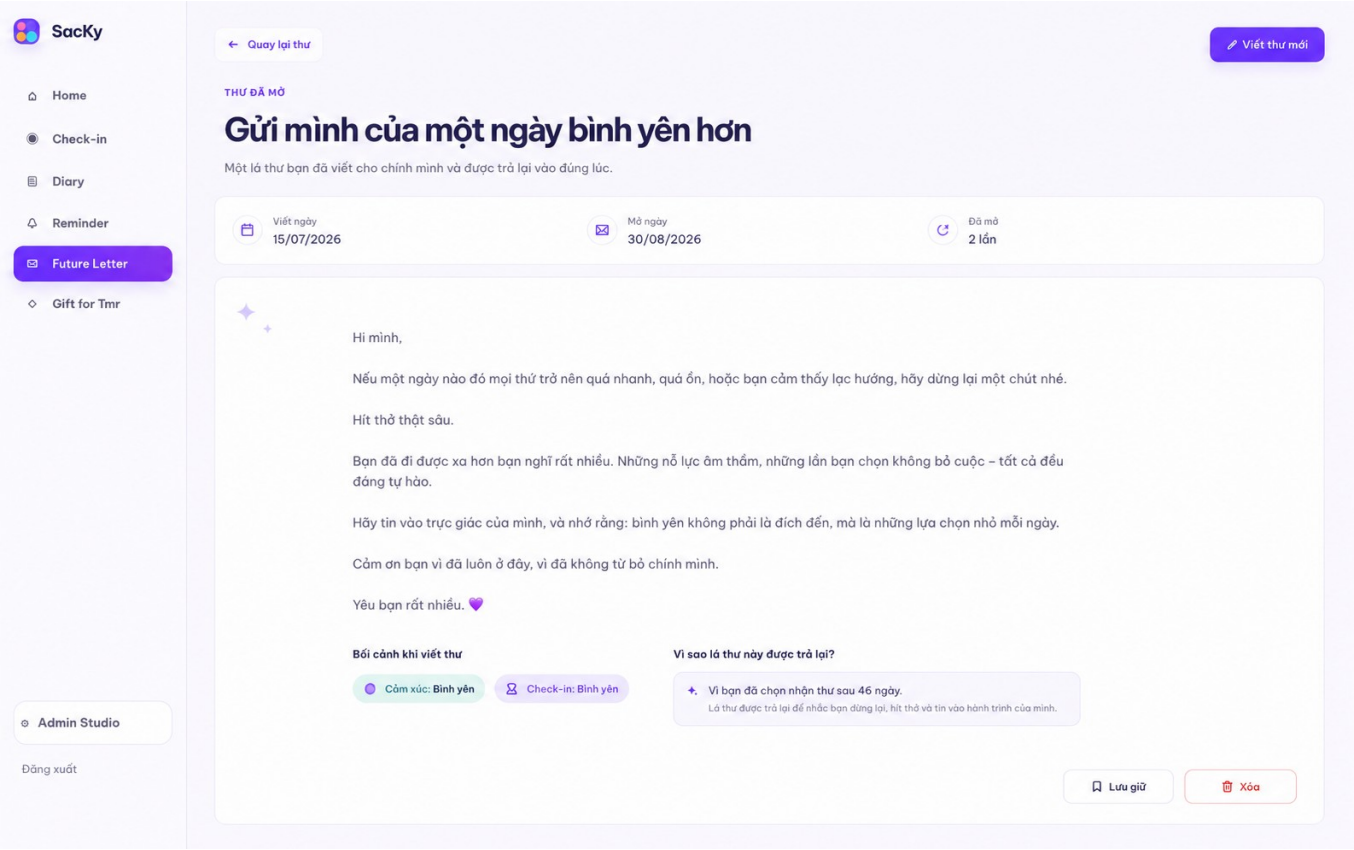


Approved desktop direction - F08 - Available Letter

Codex implementation contract

- Large envelope as primary interaction target.
- Click envelope/seal directly; no mandatory “Mở ngay” button.
- Can show why it arrived using trigger evidence.
- Other available letters shown in bounded horizontal rail/carousel.

F09 - Read Letter

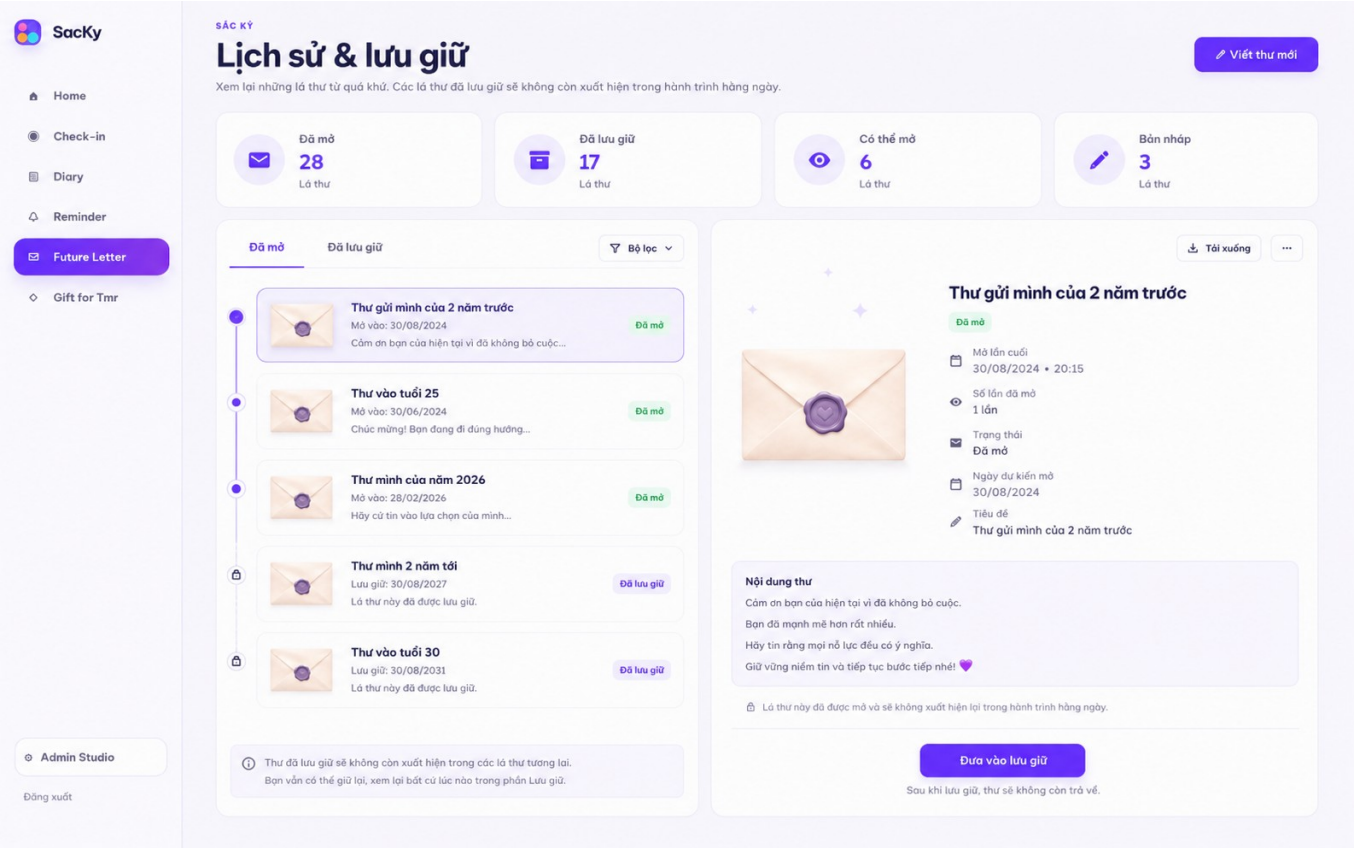


Approved desktop direction - F09 - Read Letter

Codex implementation contract

- Editorial reading surface, not email reader.
- Show written date, open history count, optional context and trigger explanation.
- Actions: Close / Archive (“Lưu giữ”) / Delete.
- No reflection.

F10 - History & Archive



Approved desktop direction - F10 - History & Archive

Codex implementation contract

- Separate Opened vs Archived semantics.
- History/detail split layout works well on desktop.
- Archived letters never re-enter trigger/random pool.
- Do not confuse archive with delete.

F11 - Opening Animation State

VISUAL STATE, NOT A SEPARATE PAGE ROUTE
Opening animation có thể là overlay/state transition nhưng phải được QA như một screen state riêng.

1. Envelope lifts a few pixels; background dims very lightly.
2. Seal pulses/shimmers once using neutral/violet lighting, not a custom letter accent.
3. Seal breaks/lifts with restrained motion.
4. Envelope flap opens.
5. Paper moves upward/reveals.
6. Transition into Read Letter without abrupt page jump.

- Normal motion target ~1.0-1.8s.
- Reduced motion: fade + reveal, no large transforms.
- No confetti, no floating hearts, no mascot, no full-screen magical particles.

13. FutureMe Module IA

```
Future Letter
├── Landing / Library
│   ├── Available
│   ├── Waiting
│   ├── Draft
│   ├── Opened / History
│   └── Archived
│
├── Create / Edit Draft
│   ├── Write
│   ├── Trigger Builder
│   ├── Decorate
│   ├── Preview
│   └── Seal
│
└── Open / Read
    ├── Available envelope
    ├── Opening animation
    ├── Read mode
    ├── Archive
    └── Delete
```

14. Service & Ownership Boundaries

- FutureMe owns letter lifecycle, trigger evaluation, random daily roll, eligibility, opening, archive/delete.
- Check-in owns Check-in data/streak/emotion records. FutureMe reads via service/repository; does not mutate Check-in.
- Diary owns Diary streak. FutureMe reads canonical metric.
- Action Space owns completion data. FutureMe reads canonical completed Action counts.
- Home only consumes FutureMe projections and invokes canonical open action.
- Admin Studio/reference layer may own future trigger config/asset catalogs; FutureMe snapshots relevant data at Seal/Trigger.

15. Runtime Evaluation Recommendations

- Daily random evaluation uses user-local calendar day and is idempotent: no more than one pool roll per local day.
- Specific-date evaluation can run daily or query-derived but must avoid duplicate trigger events.
- Streak/emotion/action conditions can evaluate on relevant domain events and/or reconciliation job.
- All evaluation paths must be idempotent.
- If runtime misses an event, reconciliation should still allow >= style conditions to trigger later where semantically valid.
- Do not rely only on client being open; eligibility should persist in backend/runtime data.

16. Edge Cases Codex Must Handle

1. No random letters in pool → no random event.
2. Random pool has letters already available/unopened → do not select those again as new events.
3. Archived random letter → excluded permanently until/unless explicit restore exists.
4. Specific date passed while user offline → becomes available on reconciliation.
5. Multiple conditional groups qualify same day → one availability event is enough; record evidence carefully without duplicating the same letter card.
6. A letter available yesterday but unopened → remains available today.
7. A letter opened today → Home persistence until day-end per projection rule.
8. Emotion-frequency chain closes without threshold → next qualifying occurrence starts next chain (recommended deterministic behavior).
9. Attached Check-in later deleted → letter must remain readable; use safe snapshot/reference fallback rather than crashing.
10. Reference Emotion renamed/deactivated → letter context and trigger history remain intelligible via snapshots.
11. Delete letter from any state → remove from UI/projections and prevent future trigger evaluation.
12. Seal action fails → remain Draft; do not show fake sealed state.

17. Acceptance Criteria - Product

☐ Draft exists and is editable.
☐ Title required; body required plain multiline.
☐ No prompts, no media, no letter color accent.
☐ Emotion/Check-in context attachments are optional and may come from any date.
☐ Authoring flow is Write → Decorate → Preview → Seal.
☐ Envelope and Seal placeholder asset selectors exist.
☐ Seal freezes content/trigger/decoration.
☐ Random pool chance is exactly modeled as 30% per local day for the pool, max 1 selected random letter/day.
☐ Opened random letters can return until Archived.
☐ Specific-date carry-over works.
☐ Check-in streak, Diary streak, Emotion frequency, Done Action conditions work.
☐ Comparators >, >=, =, <=, < work.
☐ AND/OR condition groups work.
☐ Emotion 30-day chain semantics start at first qualifying occurrence.
☐ Done Action supports lifetime and period/window modes.
☐ Trigger latch works.
☐ Trigger evidence persists.
☐ Archive keeps letter but removes future eligibility.
☐ Delete removes letter completely.
☐ Available unopened carries over.
☐ Opening increments opened_count and updates first/last opened timestamps.
☐ No reflection after open.

18. Acceptance Criteria - UI

☐ Desktop MacBook/16:10 matches Sắc Ký shell and proportions.
☐ Sidebar, typography, buttons, borders, radii, spacing align with Home/Check-in/Diary.
☐ Each canonical screen is implemented as its own clear state; do not merge mockups into one mega-dashboard.
☐ No mobile-first storyboard styling.
☐ No giant wrapper rectangles or card nesting bloat.
☐ FutureMe has one visual anchor per zone, usually envelope/seal.
☐ Create/edit surfaces preserve generous writing space.
☐ Trigger Builder is readable and structured, not a dense automation console.
☐ Decoration uses placeholders that can be replaced by real assets later.
☐ Available Letter opens by clicking envelope/seal.
☐ Opening animation feels polished and supports reduced motion.
☐ Read Letter remains calm/editorial and has no reflection UI.
☐ Archive and Delete are visually/semantically distinct.
☐ Multiple available letters use bounded carousel; no visible native scrollbar.

19. Recommended Implementation Slices

Slice 1 - Audit & semantic alignment

- Inspect existing FutureMe tables/repos/routes.

- Map legacy statuses to canonical lifecycle.
- Do not destructive-rename data without need.

Slice 2 - Draft & authoring

- Title/body validation.
- Context attachments.
- Draft persistence/autosave as appropriate.

Slice 3 - Trigger model

- Structured rules/groups/comparators.
- All six trigger types.
- Serialization + validation.

Slice 4 - Trigger runtime

- Daily random pool roll.
- Specific-date evaluator.
- Event/reconciliation for streak/emotion/action.
- Evidence + idempotency.

Slice 5 - Decoration & Seal

- Envelope/Seal placeholders.
- Preview.
- Seal snapshot + immutability.

Slice 6 - Library & projections

- Draft/Waiting/Available/Opened/Archived.
- Home projection/carry-over.
- Bounded carousel.

Slice 7 - Opening ritual

- Animation state.
- Opened counters/timestamps.
- Read view.

Slice 8 - Archive/Delete

- Archive exclusion from future eligibility.
- Delete cleanup.
- History/detail.

Slice 9 - Responsive & QA

- Desktop first.
- Intermediate/mobile adaptation after desktop acceptance.
- Accessibility/reduced motion.
- Regression tests.

20. Codex Working Rules

DO NOT COMMIT UNTIL VISUAL ACCEPTANCE
Build/test success is not enough. Codex must show authenticated browser screenshots for every core screen/state

before commit.

- Use current feature branch; do not deploy production directly.
- Keep existing business data working while migrating semantics.
- Do not invent real envelope/seal assets; implement replaceable placeholders.
- Do not simplify random semantics.
- Do not implement notification or exact-time opening unless separately requested.
- Do not add reflection after open.
- Do not expose a letter color picker.
- Run lint/tests/build.
- Add regression tests for random pool daily chance/idempotency, latch, carry-over, archive exclusion, condition comparators, 30-day emotion chain, and completion windows.
- Capture desktop authenticated screenshots: Landing, Empty, Write, Trigger Builder, Decorate, Preview, Waiting, Available, Opening state/video/GIF if tooling allows, Read, History/Archive.
- Compare screenshots against the embedded approved desktop directions and Home/Check-in/Diary styling.

21. Definition of Done

1. FutureMe behaves as a reusable future-letter system, not a one-time scheduled message.
2. Draft → Write → Decorate → Preview → Seal is complete and persistent.
3. Structured AND/OR trigger engine supports all agreed conditions and comparators.
4. Random pool semantics are correct: 30% pool/day, max one random selection/day, weighted toward lower opened_count.
5. Letters can reappear after opening until archived.
6. Archive vs Delete semantics are correct.
7. Home/Library availability and carry-over are correct.
8. Opening ritual is visually polished; no reflection is added.
9. Desktop UI matches Sắc Ký canonical style and is visually consistent with Home, Check-in and Diary.
10. All regression tests, lint and build pass, and authenticated screenshots have been reviewed before commit.

FINAL NOTE TO CODEX

When mockup text conflicts with this specification, this written Product/Runtime contract wins for behavior. For visual composition, use the embedded desktop mockups plus the canonical Sắc Ký style guide. Do not copy erroneous generated labels literally (e.g., legacy “Reminder” nav); use current product naming such as Action Space where applicable.